Supplementary Materials for Federated Learning with Stage-wise Trajectory Matching for UAV Networks on Non-IID Data

A Proof of Theorem 1 in the main paper

This supplementary material provides a detailed derivation of the convergence results stated in Theorem 1 of Section V-A. We build upon the assumptions and notation defined in the main body of the paper. To simplify the theoretical derivation, we only consider the effect of stage-wise trajectory matching while neglecting regularization terms. This does not affect the main conclusions.

When comparing stage-wise and single-stage fine-tuning processes, we note that both methods share identical optimization dynamics before the switching point t_3. Therefore, we only need to consider the process after t_3. When $t > t_3$, the server performs the E_2 fine-tuning steps using the synthetic dataset D_{syn}, then the server sends the fine-tuned global model to clients for the local training and aggregation step. Our analysis follows the process of fine-tuning, local training, and aggregation.

(1) Convergence during fine-tuning. We first analyze the convergence behavior during the fine-tuning stage on the distilled data $\mathcal{D}_{\text{syn}}$. The model update at each fine-tuning step is

$$\omega^{t,k+1} = \omega^{t,k} - \eta_{t,k} \nabla \left[\mathcal{L}\big(\omega,\, \mathcal{D}_{\text{syn}}\big) \right]_{\omega=\omega^{t,k}}, \;\; k = 0, 1, \ldots, E_2 - 1. \tag{1}$$

We define the expected error $\Delta_{t,k} = \mathbb{E}||\omega^{t,k} - \omega^*||^2$. Based on Lemma 3 in [1] and Theorem 1 in [2], the expected error satisfies

$$\Delta_{t,k+1} \leq \rho_{t,k}\, \Delta_{t,k} + \eta_{t,k}^2 B_{\text{fin}}, \tag{2}$$

where

$$\rho_{t,k} = 1 - (\mu - \delta \tilde{L})\eta_{t,k} + \mathcal{O}(\eta_{t,k}^2), \qquad B_{\text{fin}} = 2(1+\zeta)\Big(1 + \tfrac{\delta^2}{\zeta}\Big)\Upsilon^2 + (1+\zeta^{-1})\varepsilon^2. \tag{3}$$

Here, $\zeta > 0$ is a fixed positive auxiliary constant, and $\Upsilon^2 = \frac{1}{N}\sum_{i=1}^{N} \big|\big|\nabla \mathcal{L}_i(w^*, \mathcal{D}_{\text{syn}})\big|\big|^2$. For both early-stage and late-stage distilled data, Υ^2 satisfies

$$\begin{aligned}
\Upsilon^2 &= \frac{1}{N}\sum_{i=1}^{N} ||\nabla \mathcal{L}_i(w^*,\, \mathcal{D}_{\text{syn}})||^2 \\
&\leq \max_i \; ||\nabla \mathcal{L}_i(w^*,\, \mathcal{D}_{\text{syn}})||^2 \\
&= \max_i \; ||\mathbb{E}_{\xi \sim \mathcal{D}_{\text{syn}}}[\nabla \mathcal{L}_i(w^*, \xi)]||^2 \\
&\leq \max_i \; \mathbb{E}_{\xi \sim \mathcal{D}_{\text{syn}}}[\, ||\nabla \mathcal{L}_i(w^*, \xi)||^2 \,] \\
&\leq G_2^2.
\end{aligned} \tag{4}$$

By substituting the upper bound above Υ^2 into (2), the unified recursive inequality can be obtained by

$$\Delta_{t,k+1} \leq \rho_{t,k}\, \Delta_{t,k} + C_{t,k}, \tag{5}$$

where

$$C_{t,k} = \eta_{t,k}^2 \tilde{B}_{\text{fin}} = \eta_{t,k}^2 \left[2(1+\zeta)\Big(1 + \tfrac{\delta^2}{\zeta}\Big) G_2^2 + (1+\zeta^{-1})\varepsilon^2 \right].$$

Expanding the recursion in (5) across all E_2 fine-tuning steps within a communication round gives

$$\Delta_{t,E_2} \leq \left(\prod_{k=0}^{E_2-1} \rho_{t,k}\right) \Delta_{t,0} + \sum_{u=0}^{E_2-1} \left(\prod_{k=u+1}^{E_2-1} \rho_{t,k}\right) C_{t,u}. \tag{6}$$

(2) Convergence across local training and aggregation. After completing E_2 fine-tuning steps, the global model undergoes a round of local training and aggregation, which follows the FedAvg paradigm. Based on Lemma 2 in [1], the error after aggregation satisfies

$$\Delta_{t+1,0} \leq (1 - \mu\eta_t)\, \Delta_{t,E_2} + \eta_t^2 B_{\text{loc}}. \tag{7}$$

where $\Delta_{t+1,0} = \Delta_{t+1}$.

Substituting (6) into (7), we obtain the recurrence relation between communication rounds

$$\Delta_{t+1} \leq \Phi_t\, \Delta_t + Q_t + \eta_t^2 B_{\text{loc}}, \tag{8}$$

where

$$\Phi_t = (1 - \mu\eta_t) \prod_{k=0}^{E_2-1} \rho_{t,k}, \qquad Q_t = (1 - \mu\eta_t) \sum_{u=0}^{E_2-1} \left(\prod_{k=u+1}^{E_2-1} \rho_{t,k}\right) C_{t,u}. \tag{9}$$

Iterating (8) on all late-stage rounds from $t_3 + 1$ to T gives the general error bound at the end of training:

$$\Delta_T \leq \left(\prod_{t=t_3+1}^{T} \Phi_t\right) \Delta_{t_3+1} + \sum_{\tau=t_3+1}^{T} \left(\prod_{t=\tau+1}^{T} \Phi_t\right) \left(Q_\tau + \eta_\tau^2 B_{\text{loc}}\right) = \mathcal{V}_T. \tag{10}$$

The final error bound $\mathcal{V}_T$ depends on the distilled data used during fine-tuning through the parameters δ and ε embedded in $\rho_{t,k}$ and $C_{t,k}$. As justified in the main text, the late-stage distilled data $D_{\text{syn}}^{\text{late}}$ is superior to $\delta_{\text{late}} < \delta_{\text{early}}$ or $\varepsilon_{\text{late}} < \varepsilon_{\text{early}}$. This directly implies the following component-wise inequalities:

$$\rho_{t,k}^{(\text{stage})} = 1 - (\mu - \delta_{\text{late}}\tilde{L})\eta_{t,k} + \mathcal{O}(\eta_{t,k}^2) \leq 1 - (\mu - \delta_{\text{early}}\tilde{L})\eta_{t,k} + \mathcal{O}(\eta_{t,k}^2) = \rho_{t,k}^{(\text{single})}, \tag{11}$$

and

$$\begin{aligned} C_{t,k}^{(\text{stage})} &= \eta_{t,k}^2 \Big[2(1+\zeta)\Big(1 + \tfrac{\delta_{\text{late}}^2}{\zeta}\Big) G_2^2 + (1+\zeta^{-1})\varepsilon_{\text{late}}^2 \Big] \\ &\leq \eta_{t,k}^2 \Big[2(1+\zeta)\Big(1 + \tfrac{\delta_{\text{early}}^2}{\zeta}\Big) G_2^2 + (1+\zeta^{-1})\varepsilon_{\text{early}}^2 \Big] = C_{t,k}^{(\text{single})}, \end{aligned} \tag{12}$$

with at least one of the above inequalities being strict.

Consequently, the coefficients defined in (8) satisfy

$$\Phi_t^{(\text{stage})} = (1 - \mu\eta_t) \prod_{k=0}^{E_2-1} \rho_{t,k}^{(\text{stage})} \leq (1 - \mu\eta_t) \prod_{k=0}^{E_2-1} \rho_{t,k}^{(\text{single})} = \Phi_t^{(\text{single})}, \tag{13}$$

and

$$Q_t^{(\text{stage})} = (1-\mu\eta_t) \sum_{u=0}^{E_2-1} \left(\prod_{k=u+1}^{E_2-1} \rho_{t,k}^{(\text{stage})}\right) C_{t,u}^{(\text{stage})} < (1-\mu\eta_t) \sum_{u=0}^{E_2-1} \left(\prod_{k=u+1}^{E_2-1} \rho_{t,k}^{(\text{single})}\right) C_{t,u}^{(\text{single})} = Q_t^{(\text{single})}. \tag{14}$$

Substituting these relations into the expression for $\mathcal{V}_T$ in (10) leads to the final comparison obtained by

$$\mathcal{V}_T^{(\text{stage})} < \mathcal{V}_T^{(\text{single})}. \tag{15}$$

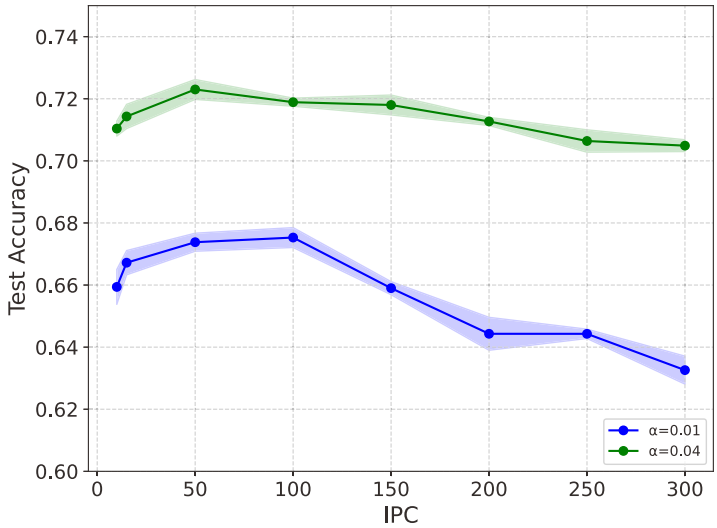


Fig. 1: Final test accuracies of FedSTM with different numbers of images per class (IPC) in the late-stage distilled data.

This completes the proof, showing that the stage-wise strategy achieves a strictly tighter upper bound on the expected error than the single-stage approach.

B Impact of different distilled data sizes

We evaluated the performance of FedSTM under different sizes of late-stage distilled data. This experiment was carried out on the CIFAR10 dataset. Fig. 1 illustrates the final test accuracies of FedSTM with different numbers of images per class (IPC) in the distilled data in the late stage under two levels of data heterogeneity: $\alpha = 0.01$ and $\alpha = 0.04$.

The results indicate that the performance of the global model first improves and subsequently declines as the size of the late-stage distilled data increases. When the distilled data size is small, the limited number of samples makes it difficult for the model to learn complex patterns in the global data distribution, thereby restricting its performance. As the size of the distilled data increases, the performance of the global model improves significantly. However, beyond a certain threshold, the performance gain plateaus, showing only marginal improvements. When the distilled data size per class becomes excessively large, the global model's accuracy drops noticeably. This is because the distilled data are randomly initialized with Gaussian noise, and an overly large size leads to significant noise remaining in the distilled data. This noise negatively impacts the fine-tuning of the global model. Therefore, in practical applications, it is crucial to carefully control the size of the distilled data to achieve optimal performance.

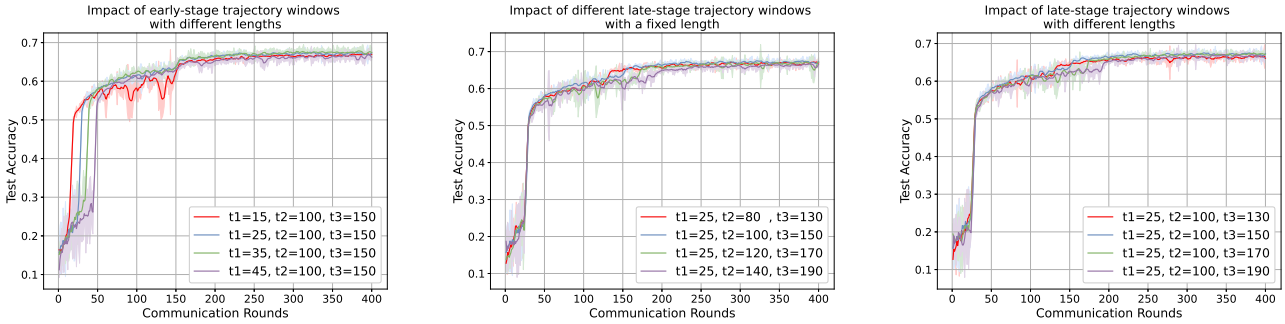
C Impact of different starting and ending rounds for early-stage and late-stage trajectory matching

To evaluate the impact of different starting and ending rounds for early-stage and late-stage trajectory matching, We first fix the values of t_2, and t_3, and change the ending round t_1 of the early-stage trajectory. Fig. 2a illustrates the test precision of the global model with different lengths of the early-stage trajectory on CIFAR

10 when $\alpha = 0.01$. We can see that adjusting the value of t_1 significantly affects the convergence speed of the global model. A smaller t_1 makes the global model fine-tune on the distilled data earlier, thus accelerating the convergence process. Meanwhile, the final performance of the model is almost unaffected, indicating that the distilled data can effectively extract common features even from a shorter early-stage trajectory. However, an excessively short early-stage trajectory, i.e., $t_1 = 15$, can result in insufficient effective information captured by the distilled data, leading to oscillations in the convergence of the model.

Then we fix the value of t_1 and investigate the impact of t_2 and t_3. Fig. 2b illustrates the test accuracy of the global model with fixed length of the late-stage trajectory but different starting rounds and ending rounds. We can see that advancing the late-stage trajectory matching can further accelerate the model's convergence process in the later stages of training, while the final performance remains unaffected. Subsequently, we fix the values of t_1 and t_2, and modify only the final round t_3 of the late-stage trajectory, thus changing the length of the late-stage trajectory. As shown in Fig. 2c, even with a significantly longer late-stage trajectory, the final performance remains essentially stable. This indicates that the distilled data synthesized from a shorter late-stage trajectory is already sufficient to effectively capture the fine-grained features present in the later stages of training.

A comprehensive analysis of the above experimental results reveals that variations in the parameters t_1, t_2, t_3 only affect the convergence rate of the global model and have a minor impact on the final performance. This fully demonstrates that the proposed method exhibits strong robustness with respect to both the trajectory rounds and the trajectory length.



(a) (b) (c)

Fig. 2: Impact with different starting and ending rounds for early-stage and late-stage trajectory matching.

REFERENCES

[1] R. Pi, W. Zhang, Y. Xie, J. Gao, X. Wang, S. Kim, and Q. Chen, "Dynafed: Tackling client data heterogeneity with global dynamics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 12 177–12 186.

[2] B. Hu, P. Seiler, and L. Lessard, "Analysis of biased stochastic gradient descent using sequential semidefinite programs," *Math. Program.*, vol. 187, no. 1, pp. 383–408, 2021.